

Skills Portfolio

BEN MANSOUR MANSOUR

Full Stack Engineer / Ingénieur Full Stack

Introduction

My name is **Mansour BENMANSOUR**, a passionate **Full Stack Engineer** specializing in building scalable, high-performance web applications. I hold a degree in **Multimedia and Web Development** from **ESPRIT (École Sup Privée d'Ingénierie et de Technologies)**, where I graduated with **high distinction** in 2023. Since then, I have gained over **2 years of professional experience** across both remote and on-site environments, working with modern stacks such as **Next.js, Node.js, GraphQL, Supabase, and SQL Server**. My goal is to deliver impactful digital solutions that align with business needs while maintaining clean architecture, performance, and long-term scalability.

Skills

- **Programming Languages:** JavaScript, TypeScript, Python, SQL, Java, Bash
- **Backend Frameworks:** Node.js, Express, Django, Flask, Spring Boot
- **Frontend Frameworks:** React, Next.js, Angular, Vue.js
- **Databases:** MongoDB, PostgreSQL, MySQL, Redis
- **Development Tools:** Git, Docker, Nginx, CloudFlare Webpack, GraphQL, Jest, AWS, Kubernetes, CI/CD pipelines (GitHub Actions, Jenkins)
- **APIs and Payment Integrations:** REST APIs, PayPal, Stripe, APS, NextAuth, OAuth
- **CSS Frameworks:** Tailwind CSS, Bootstrap, Material-UI
- **Other Skills:** Elasticsearch, RabbitMQ, WebSockets, Microservices architecture

Professional Experience

Full Stack Engineer & Infrastructure Architect — Edifis Revendeurs Hub

Collaborative Project — Technical Deliverables

- **Multi-Site E-Commerce Engine (Shopify-Style):** Architected a specialized B2B multi-tenant e-commerce platform dedicated to industrial sanitary equipment parts, serving a growing user base of **1,000+ active operators** and managing tens of thousands of stock-keeping units (SKUs).
- **Dynamic Website Provisioning Core:** Developed a state-intensive, centralized Administration Portal that allows administrators to instantly spawn, brand, and configure fully independent reseller storefronts. Engineered the underlying meta-programming layers to dynamically map specific tenant parameters into modular, consumer-facing frontends.
- **Edge Caching & Live Storefront Routing:** Deployed and optimized the system's global network routing topology via [Cloudflare](#). Structured dynamic reverse-proxy rules, isolated sub-domain handlers, and custom routing configurations to seamlessly serve live reseller environments across the ecosystem (e.g., *ekm.sanitaire.shop*, *sfc.sanitaire.shop*) with sub-second latency. (+60 website)
- **Automated ERP & E-Commerce Synchronization:** Designed and implemented data-sync pipelines to bridge internal infrastructure gaps. Engineered robust workers that reliably automate real-time synchronization of complex multi-tier inventory levels, live retail margins, and price changes between corporate systems (**Sage**), external storefront endpoints (**Ecwid**), and the primary database layer.
- **State Management Architecture Refactoring:** Successfully eliminated local storage settings anti-patterns across user dashboards. Standardized client configuration states with a unified state-management architecture, enabling smooth on-the-fly transformations between heavy spreadsheet-style data grids and product card matrices without client-side lag.
- **Database Integrity & Sandbox Isolation:** Overhauled isolated SQL schema scripts, explicitly fixing identity insertion behaviors and deep multi-table foreign key dependencies. Guaranteed secure database transactions and zero-error deployments across unique single-tenant database instances during new store rollouts.

Technical Lead & Solutions Architect — Admission ERP Platform

June 2025 – Present — Tunis, Tunisia

- **Enterprise Core Architecture:** Orchestrated the design and end-to-end implementation of a highly scalable, multi-center ERP system centralizing institutional operations across three major regional branches (Tunis, Sfax, Sousse). Built the client layer on [Next.js 15 \(App Router\)](#) utilizing React Server Components, communicating with a decoupled [Node.js / TypeScript](#) microservices backend engineered for high concurrency.
- **Relational Data Modeling & Storage:** Architected a mission-critical database schema on [Microsoft SQL Server](#) to ensure transactional integrity across multi-table operations. Optimized stored procedures, relational mappings, complex query joins, and custom identity constraints to maintain sub-second server response times under intense peak registration loads.
- **Academic Scheduling & Presence Engines:** Conceived and developed state-heavy custom workflow engines to automate complex course scenarios. This includes automated timetabling, professor allocation algorithms, session conflicts management, and a real-time, tamper-proof student presence tracking dashboard.
- **International Certification Pipelines:** Engineered data compliance and automated generation modules dedicated to strict European certification standards (**ÖSD** and **telc** language exams). Built secure document export frameworks capable of dynamically compiling, parsing, and printing legal student credentials while maintaining rigorous data trace-history.
- **Financial Ledger Subsystem:** Developed a secure internal financial auditing layer to manage multi-tier pricing structures, installment payment workflows, invoice generation, and balance reconciliation. Implemented strict data isolation layers to safeguard cashflow records and historical financial ledgers.
- **DevOps, Clustering & Process Monitoring:** Established a high-availability production infrastructure utilizing [PM2](#) process management, process cluster-toggling, and automated failure-recovery configurations. Configured detailed runtime logging mechanisms to instantly capture backend bottlenecks, minimizing system downtime to near-zero.
- **Deployment Pipelines & Git Workflows:** Structured corporate automation pathways by enforcing a strict Gitflow version-control strategy on [GitHub](#). Supervised automated checking, testing, and production environment separation layers while overseeing pull-request validations to protect main-branch stability.
- **Change Management & High-Scale Onboarding:** Acted as the primary technical owner for full product rollout. Designed extensive technical reference manuals and personally spearheaded intensive onboarding training workshops for over **60+ active corporate users**, achieving frictionless user migration from legacy tools.

Full Stack Developer (DevOps Focus) - BeBouncy

February 2024 – April 2025 — La Marsa, Tunisia **Project:** Social Media Management Platform — **Team:** 10-member cross-functional team

- **Overview of the platform:** Developed comprehensive solution enabling content creators to:
 - Manage posts/comments/likes across Facebook, Instagram, YouTube, LinkedIn
 - Track performance analytics for multiple connected pages
 - Schedule and bulk-publish content with approval workflows
- **DevOps Transformation:**
 - Modernized infrastructure from manual SSH to fully automated deployments
 - Implemented Docker-based isolation for social media API integrations
 - Developed rate-limiting system to prevent platform blacklisting
- **Key Technical Achievements:**
 - **CI/CD Pipeline:** Reduced deployment time from 15min → 3min (80% improvement)
 - **Multi-Platform Integration:** Unified API layer for 4 social networks
 - **Data Processing:** Handled 50K+ daily events with Redis caching
- **Core Technical Stack:**
 - **Frontend:** Next.js 14, Redux Toolkit, Chart.js
 - **Backend:** Node.js 18, Express, BullMQ
 - **Infrastructure:** Docker, Portainer, GitHub Actions
- **Business Impact:**
 - Enabled centralized management of 5+ social accounts
 - Reduced content scheduling time by 60% through batch processing
 - Implemented compliance features for platform API usage limits
- **Technical Leadership:**
 - Upgraded Node.js 12 → 18 and Next.js 11 → 14.5 with zero downtime
 - Implemented Stripe payment processing (one-time + subscriptions)
 - Identified 40% performance optimization opportunities through audits
- **Frontend Development:**
 - Built Next.js frontend with Redux state management
 - Optimized production builds using Webpack
 - Reduced bundle size by 35% via code splitting
- **Cross-Team Collaboration:**
 - Translated marketing requirements into technical specifications
 - Worked with designers on UI implementation (Figma → React)
 - Mentored junior developers on Git workflows and deployment procedures

Sikka Software

Full Stack Developer — August 2022 – Sep 2025 — Remote

Tanad.app - Plateforme ERP/Facturation Entreprise Complète

tanad.app

Mai 2025 – Feb 2026

- **Architecture Technique & Innovation:** Conception d'une solution ERP modulaire et complète, spécialement adaptée au marché saoudien :
 - **Design modulaire avancé** : Système d'activation/désactivation de modules (40+ modules) permettant une personnalisation complète selon les besoins métier
 - **Architecture multi-locataire** avec isolation des données via Row Level Security (RLS) [PostgreSQL](#)
 - **Synchronisation temps réel** entre multiples entrepôts et branches via [WebSocket](#) et Supabase Realtime
 - **Journalisation complète** de toutes les transactions financières et mouvements de stock avec audit trail
- **Modules Fonctionnels Développés:**
 - **Gestion Financière** : Système de facturation électronique, devis, comptes clients/fournisseurs
 - **Ressources Humaines** : Gestion employés, recrutement, paie, départements et postes
 - **Gestion Opérationnelle** : Multi-entrepôts, catalogue produits, achats, véhicules
 - **Administration** : Multi-succursales, rôles/permissions, actifs fixes, analytiques
- **Conformité Réglementaire ZATCA:** Développement d'une solution conforme à la réglementation saoudienne incluant :
 - Implémentation complète des standards [UBL 2.1](#) pour ZATCA Phase 2
 - **Horodatage cryptographique** des factures répondant aux exigences FATCA
 - **Reporting automatisé** vers le portail ZATCA via API gouvernementales
 - **Génération de codes QR TLV** contenant toutes les informations fiscales requises
 - Validation automatique de la conformité des factures
- **Stack Technique Moderne:**
 - Frontend développé avec [Next.js 15](#) utilisant l'architecture App Router et [TypeScript](#) strict
 - Backend : API RESTful + [GraphQL](#) fédéré, [Node.js](#) avec microservices
 - Base de données : [PostgreSQL](#) avec Supabase, politiques de sécurité au niveau des lignes
 - Déploiement : Containerisation [Docker](#), déploiement via Coolify sur VPS
- **Innovations Techniques:**
 - **Performance & Scalabilité** : Architecture microservices, cache intelligent, notifications temps réel
 - **Sécurité** : Chiffrement bout en bout, MFA, audit trail complet, conformité standards internationaux
 - **Interface multilingue** (Arabe/Anglais) adaptée au marché GCC

Pukla (All-in-One Link Management Platform)

puk.la

Sep 2023 – Feb 2024

- **Core Platform Features:** Developed a comprehensive link management solution with tiered functionality:
 - **Free Tier:**
 - * Basic link management with drag-and-drop organization
 - * Standard analytics (click counts, referrers)
 - * Limited theme customization options
 - **Pro Tier ([Stripe](#) Subscription):**
 - * Advanced analytics with geographic heatmaps and device breakdowns
 - * Custom domain support with SSL certification
 - * Priority support and early access to new features
 - **Enterprise Tier:**
 - * White-label solutions with custom CSS/JS injection
 - * Team collaboration features with role-based permissions
 - * API access for programmatic link management
- **Payment System:**
 - Implemented [Stripe](#) integration supporting:
 - * Recurring subscriptions with prorated upgrades/downgrades
 - * Multiple payment methods (credit cards, Apple Pay, Google Pay)
 - * Automated invoice generation and dunning management
 - Custom billing portal for plan management and payment history
 - Webhook handlers for subscription lifecycle events
- **Technical Architecture:**
 - Monorepo structure using [PNPM workspaces](#) with:
 - * Shared TypeScript configuration across packages
 - * Dedicated packages for UI components, utilities, and API types
 - * Isolated test environments for each subsystem
 - Frontend built with:
 - * [Next.js 14](#) with App Router for file-based routing
 - * [shadcn/ui](#) component library with Radix primitives
 - * [Framer Motion](#) for performant animations and micro-interactions
 - Backend services:
 - * [Supabase](#) PostgreSQL with row-level security
 - * Redis caching for high-traffic endpoints
 - * Rate limiting and abuse protection
- **Design System:**
 - Micro-interactions throughout UI:
 - * Smooth transitions between route changes
 - * Hover effects on interactive elements

- * Loading skeletons for async content
- Custom animation presets for:
 - * Modal appearances/disappearances
 - * Form submission states
 - * Notification toasts
- Responsive design supporting:
 - * Mobile-first layout approach
 - * Adaptive component sizing
 - * Reduced motion preferences
- **Developer Experience:**
 - Comprehensive development tooling:
 - * Hot module replacement for immediate feedback
 - * Visual regression testing setup
 - * Storybook documentation for UI components
 - Deployment infrastructure:
 - * Preview deployments for every Git branch
 - * Automated database migrations
 - * Rollback capabilities for failed deployments

Laz.im (Enterprise URL Shortener)

laz.im

- **Technical Architecture:**

- [Next.js 14](#) (App Router) with:
 - * [shadcn/ui](#) component library
 - * Edge Runtime for global redirects
 - * Incremental Static Regeneration for dashboards
- [Supabase](#) backend featuring:
 - * PostgreSQL with row-level security
 - * Realtime subscriptions for analytics
 - * Built-in authentication

- **Payment System:**

- [Stripe](#) integration handling:
 - * Usage-based billing (clicks/month)
 - * Team seat management
 - * [Subscription](#) webhooks with Supabase Functions
- PCI-compliant billing portal
- Promo code system with referral tracking

- **Core Features:**

- Real-time analytics dashboard:
 - * Geographic heatmaps
 - * [Device](#) and browser breakdowns
 - * UTM parameter tracking
- Custom domains with SSL
- API access with rate limiting

- **Developer Experience:**

- Monorepo with [PNPM workspaces](#)
- Automated testing pipeline
- Preview deployments for all branches

Payment Orchestration Platform

- **Payment Processing:** Created a unified payment hub supporting:
 - Multiple gateways including [Stripe](#), [PayPal](#), and regional providers
 - Both one-time and recurring payments
 - Automated reconciliation of bank transfers
 - Smart routing to optimize payment success rates
 - Seamless accounting system integration
- **Technical Architecture:**
 - [GraphQL](#) API gateway abstracting provider differences
 - Idempotent transaction processing with retry logic
 - Comprehensive webhook handling system
 - Subscription lifecycle management
- **Security Measures:**
 - PCI DSS compliant architecture with encrypted sensitive data
 - JWT token rotation and strict session management
 - Regular third-party security audits
 - SOC 2 Type II certified infrastructure

Education

Engineering in Computer Science (TWIN)

Esprit, Ariana - El Ghazala

2020 – 2023

- Graduated with a focus on **Full Stack Development**, including extensive training in both **front-end** and **back-end** technologies.
 - Engaged in various projects, emphasizing practical applications of **computer science** theories in real-world scenarios.
 - Developed skills in **team collaboration**, **project management**, and **software development lifecycle**.
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Degree in Multimedia and Web Development

Institut Supérieur des Etudes Technologiques de Djerba, Iset Djerba - Midoun

2017 – 2020

- Acquired foundational knowledge in **web technologies**, **design principles**, and **multimedia applications**.
- Worked on several projects that integrated **multimedia elements** into **web applications**, enhancing user experience.
- Participated in **workshops** and **training sessions** focused on the latest trends in **web development** and **design**.

1 Papiers / Documents

Certificate of Employment



Certificate of Employment

This is to certify that

Mansour BenMansour

Has been employed in

Sikka Software

(مؤسسة سكة الأفكار لتقنية المعلومات)

As

Full-stack Developer

From September 2022 up to the present

This certification is being issued upon the request of the aforementioned name
for whatever lawful purpose it may serve.

Issued on this

28th day of May 2025

Zakher M. Masri

Chief Executive Officer
Sikka Software



Issued by Esprit, 2023



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